

RMDL Assessment 2: Data Analysis Report.

Emotional Stroop effects in French-English bilingual speakers

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0.1 Background

The Stroop effect is a cognitive interference effect from the Stroop task (Stroop 1935), whereby identification of a written colour word's ink colour is slower when the written word is printed in a conflicting colour: for example, the word 'blue' printed in green ink is more slowly identified than when it is printed in blue ink, due to interference from the automatic process of reading during colour identification. Subsequent research has led to several variations of the Stroop task. One such variant is the emotional Stroop task, in which participants identify the ink colour of emotionally salient and neutral words rather than colour words. The resulting interference effect is theorised to be the result of interference caused by the emotion stimuli, the process for which is automatic and immediate [Bargh (1997)]. In addition, the Stroop paradigm has been widely applied in bilingual contexts to examine patterns of interference across languages [Mägiste (1984)]. In bilinguals, research often reflects that the mother tongue (L1) is more strongly associated with emotion, while the second language (L2) can be used to create emotional distance (Dewaele 2010). Sutton et al., (2007) investigated the bilingual emotional Stroop task in Spanish bilinguals. They found an emotional Stroop effect, with highly efficient bilinguals displaying similar emotional interference effect in both languages, and concluded that the results reflect that emotional valence of lexical items can be automatically activated in more than one language. Sutton et al., (2007) had one participant group of English-Spanish early bilinguals, therefore the present study builds on the findings to investigate to what extent age of acquisition influences the emotional interference effect observed in an L1 and L2, in French-English bilingual speakers. The following study builds on the theory of the emotional Stroop effect and bilingual Stroop effect, to investigate the effect of age of acquisition of emotional interference in English-French bilingual speakers.

0.1.1 Research Questions/Hypotheses

Research question:

The research question asks broadly if age of acquisition influences reaction time to French and English emotion compared to neutral words in a bilingual Stroop paradigm. Specifically, the present study investigates if an emotional interference effect is observed similarly or differentially across early and late bilinguals groups in an emotional bilingual Stroop paradigm across languages (French compared to English).

Research hypotheses: Hypotheses is based on the results of Sutton et al. (2007).

- **Hypothesis 1:** Overall, we can predict that both groups will show a slower reaction time in the emotion condition compared to the neutral condition, due to the emotional interference effect.
- **Hypothesis 2:** In early bilinguals, since they have fluency in both French and English and acquired both languages in early childhood, it is predicted that this group will display an equal amount of emotional interference across both languages, meaning similar reaction times will be shown.
- **Hypothesis 3:** For late bilinguals, a slower reaction time is predicted for emotion words than for neutral words only in the English condition, since the emotional interference effect is predicted to be stronger in the L1.

Overall, we can expect the emotional interference effect to negatively correlate with age of acquisition, such that early bilinguals will show faster reaction times than late bilinguals to French emotional stimuli.

0.2 Methods

0.2.1 Participants

The study uses two participants groups: early sequential and simultaneous bilingual French-English speakers (attainment of both languages before the age of 11) and late sequential French-English speakers (native English speakers who learnt French as a second language after the age of 11). The late bilingual group had 6 participants, and the early bilingual groups had 5 participants, so a total of 11 participants took part in the study.

In the late bilingual group, acquisition of French was between the ages of 11 - 15 (mean = 11), with fluency (mean = 71% range = 50 - 90%) achieved from the ages of 16 - 20 (mean = 18), with one participant describing themselves as not fluent in French. All participants had proficiency and/or fluency additionally in Italian, Spanish, German, and BSL or a combination of these languages. 1 participant had acquired Italian and English from birth, and had acquired L4 French after L3 German, while the remaining 5 participants had acquired French as an L2. On average, participants were exposed to English 79.5% of the time, French 8.4% of the time, and other languages 12.1% of the time. 4 participants reported that they would choose to read

in English over another language, while 1 participant reported a 50% preference for French and English, and 1 participant reported that they would choose to read in Italian.

In the early bilingual group, 4 participants were simultaneous learners of French and English, meaning they acquired them simultaneously from birth, while 1 participant acquired French from birth and English from age 4, and 1 participant was a native French speaker who acquired fluency in English from age 10. All participants had proficiency and/or fluency additionally in Spanish, German and Russian, or a combination of these languages. On average, participants were exposed to English 73.3% and to French 23% of the time, and to another language 3.7% of the time. 4 participants reported that they would choose to read in English, while 1 reported that they would choose to read in French.

All participants were aged between 19 - 28 and are university educated, and currently reside in the United Kingdom or the United States.

0.2.1.1 Table 1 - Participant characteristics

Participant group	Age of acquisition of French (mean, range)	Self-rated fluency level in French (mean, range)	Age at which fluency in French achieved (mean, range)	Current language exposure
Late bilinguals	11, 11 - 15	71%, 50 - 90%	18, 16 - 20	English = 79.5% French = 8.4% Other = 12.1%
Early bilinguals	1, 0-4	100%	N/A	English = 73.3% French = 23% Other = 3.7%

0.2.2 Materials

Language history questionnaire

Firstly, all participants completed a language history questionnaire (Marian et al., 2007), which collects information on their language acquisition background and self-proficiency ratings. The information is presented below in Table 1 (above).

Stroop task

The Stroop task assessment was built using the materials adapted from Sutton et al. (2007). In the original study, the word list was composed of two different types of words: negative emotion words (e.g. anger) and neutral (non-emotion related) words which belonged to the category ‘nautical nouns’ (e.g. boat). The English words were repeated in Spanish. Sutton et al. (2007) highlight that the chosen words were high in arousal and low in valence, and matched for frequency and length. Therefore, the current study uses the same English word list but with French translated equivalents (rather than Spanish) in order to build on the results of Sutton et al. (2007) and investigate a different language combination. The complete word list (French & English, emotion and neutral items) was repeated in blue ink and in green ink. The word list was therefore divided into three blocks: language (French/English), word type/condition (neutral/emotion) and ink colour (blue/green). Due to the small scale of the study, the translated French words were not matched for frequency and length.

Participants saw a total of 32 items, as items were counterbalanced across colours and languages: half of the emotion items appeared in English and half in French, and the same applied to neutral items. The items were not repeated across languages (for example, if a participant saw the English word ‘boat’ they would not see the French equivalent ‘bateau’). Items were also counterbalanced across colour: each word was presented once, in one colour, and not repeated in the second colour. In summary, participants saw eight neutral English items (4 in green, 4 in blue), eight emotion English items (4 in green, 4 in blue), eight neutral French items (4 in green, 4 in blue), and eight emotion French items (4 in green, 4 in blue). Items were blocked by language, and the order that the blocks were presented in was pseudorandomised and counterbalanced across participants.

A complete list of stimuli can be found in the Appendix.

0.2.3 Procedure

The Stroop task was self-conducted online via Testable, with each participant completing it on their own device. Participants first completed a practice session, consisting of a total of 6 neutral items, which were in a distinct semantic category to the critical trial neutral items. In the practice condition, the word appeared on the screen for 3 seconds. Participants were instructed to press the ‘L’ key if the word appeared in green ink, and the ‘A’ key if the word appeared in blue ink. Instructions only appeared in English. For the practice trials, the key/colour reminder prompt appeared on the screen alongside the practice word. Participants were not given feedback on the practice condition if their answer was correct. Following the practice condition, the participants immediately completed the critical trial conditions. In the critical condition, the stimuli appeared on the screen for less time - a total of 1.7 seconds - and the prompt did not appear on the screen alongside the stimuli.

0.2.4 Statistical Analysis

In order to answer the research question, *if age of acquisition influences reaction time to French and English emotion compared to neutral words in a bilingual Stroop paradigm*, age of acquisition is treated as a categorical variable of early bilinguals and late bilinguals, and the bilingual nature of the Stroop paradigm is operationalised as language of stimuli (English, French).

Reaction times in milliseconds for each trial will be analysed using mixed-effects regression modelling. Summary measures for mean and standard deviation of the raw data will be reported. The study will then employ a 2x2x2 mixed factorial design of lognormal regression, with group (early vs late), language (French vs English) and Condition (emotion vs neutral) and their interactions as between-participant fixed effects and condition (emotion vs neutral) and language of stimuli (French vs English) as within-participant varying slopes, as well as by-subject varying intercept.

0.2.5 Data Availability

Anonymised participant data (Stroop task performance) is available on the following GitHub link: <https://github.com/KatiePedrotti/rmdl-template>

To respect participant privacy, language questionnaires are not available.

0.3 Results: Data analysis

0.3.1 Reading and processing the data

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(brms)
```

Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing `help('brms')`. A more detailed introduction
to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
library(dplyr)
library(posterior)
```

This is posterior version 1.6.1

Attaching package: 'posterior'

The following objects are masked from 'package:stats':

mad, sd, var

The following objects are masked from 'package:base':

%in%, match

```
library(tidybayes)
```

Warning: package 'tidybayes' was built under R version 4.5.3

Attaching package: 'tidybayes'

The following objects are masked from 'package:brms':

dstudent_t, pstudent_t, qstudent_t, rstudent_t

```

alldata = list.files(
  path = "./data/",
  pattern = "*.csv$",
  recursive = TRUE ,
  full.names = TRUE
)

compiledlist <- lapply(alldata, function(x) {
  read_csv(x, skip = 3)
})

```

Rows: 40 Columns: 26

-- Column specification -----

Delimiter: ","

chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...

dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Rows: 40 Columns: 26

-- Column specification -----

Delimiter: ","

chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...

dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

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```

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Delimiter: ","
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i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

```



```
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 40 Columns: 26
-- Column specification -----
Delimiter: ","
chr (12): type, condition1, stim1, textColor, stimFormat, subjectGroup, keyb...
dbl (14): id, rowNo, ITI, presTime, random, ITI_ms, ITI_f, ITI_fDuration, pr...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
finaldata <- bind_rows(compiledlist) |>
  filter(condition1 != "practise") |>
  mutate(
    early_late = case_when(
      substring(id,1,1) == "1"~"early",
      substring(id,1,1) == "2"~"late",
      TRUE ~""
    )
  )
```

As the data were in separate files and lacking identifying columns, I firstly used `list.files()` to list all the available results files, and `lapply()` to compile them together, with `read_csv(x, skip = 3)` in order to skip the first 3 rows of each dataset which did not contain any data. Then I used `bind_rows()` to read in this data into one dataset, filtered out the 6 practise conditions for each participant using `filter()`, and used `mutate()` in order to create a column which classified the participants as early or late bilinguals based on their participant number (participants numbers starting with 1 were early bilinguals, and starting with 2 were late bilinguals).

```
newfinaldata <- finaldata |>
  separate_wider_delim(
    condition1,
    delim = "_",
    names = c("Condition", "Language")
  )
```

```
newfinaldata$Language <- recode(newfinaldata$Language,
                                "eng" = "English",
                                "fr"  = "French")

newfinaldata$early_late <- recode(newfinaldata$early_late,
                                  "early" = "Early bilingual",
                                  "late"  = "Late bilingual")

newfinaldata$Condition <- recode(newfinaldata$Condition,
                                  "emotion" = "Emotion",
                                  "neutral"  = "Neutral")
```

I used `separate_wider_delim()` in order to create new columns which identified the condition (emotion and neutral) and language (French and English) (this information had been presented together in one column in the original Testable file, such as `emotion_fr`), and re-named the column names accordingly. The relevant columns are therefore `Language` (French/English stimuli), `Early_Late` (early or late bilingual), and `Condition` (emotion/neutral stimuli). Importantly, Reaction time (RT) appears in a column, which was captured by Testable.

0.3.2 Summary of the raw data

```
newfinaldata |>
  group_by(Condition, Language, early_late) |>
  summarise(
    mean = mean(RT), sd = sd(RT))
```

``summarise()`` has grouped output by 'Condition', 'Language'. You can override using the ``.groups`` argument.

```
# A tibble: 8 x 5
# Groups:   Condition, Language [4]
  Condition Language early_late    mean    sd
  <chr>      <chr>    <chr>    <dbl> <dbl>
1 Emotion   English   Early bilingual  818.  561.
2 Emotion   English   Late bilingual  542.  255.
3 Emotion   French    Early bilingual  726.  440.
4 Emotion   French    Late bilingual  533.  216.
5 Neutral   English   Early bilingual  806.  537.
6 Neutral   English   Late bilingual  516.  196.
```

7 Neutral	French	Early bilingual	756.	377.
8 Neutral	French	Late bilingual	537.	204.

In order to investigate if age of acquisition affects reaction time to emotion and neutral words in the bilingual Stroop task, I analysed participant reaction times across condition (neutral vs. emotion) and language (French vs. English) and obtained mean and standard deviation.

Across conditions, the raw data shows that late bilinguals showed faster mean reaction times than early bilinguals. The results are summarised below and displayed in Figure 1 (below).

In the emotion condition, early bilinguals showed a faster reaction time in French (mean = 726.38ms, SD = 440.26) than in English (mean = 817.83ms, SD = 561.09). This is contrary to the hypothesis, the early bilinguals would perform similarly across languages. The late bilinguals showed the same pattern: faster reaction times in French (mean = 533.33ms, SD = 215.51) than in English (mean = 542.31ms, SD = 255.28). While both groups displayed the same performance pattern, overall in the emotion condition, the late bilinguals had faster reaction times in both languages than the early bilinguals, but the early bilinguals were considerably slower in English, while the variation between language for the late bilinguals was not as pronounced. This is contrary to the hypothesis that early bilinguals will show a faster reaction time than late bilinguals in the emotion condition.

In the neutral condition, early bilinguals had a faster reaction time in French (mean = 755.85ms, SD = 376.74ms) than in English (mean = 805.58ms, SD = 536.91ms) in the neutral condition, while late bilinguals showed the reverse pattern, with a faster reaction time in English (mean = 515.77ms, SD = 196.49ms) than in French (mean = 536.77ms, SD = 204.00ms) in the neutral condition. Therefore overall in the neutral condition, late bilinguals showed faster reaction times in both languages compared to early bilinguals, but early bilinguals performed faster in French and late bilinguals performed faster in English.

Comparing conditions, early bilinguals showed a slightly faster reaction time in the emotion condition in English (English: mean = 817.83ms, SD = 561.08ms) compared to the neutral condition in English (English: mean = 805.58ms, SD = 536.91ms) but showed the reverse pattern in the emotion condition in French (emotion: mean = 726.38ms, SD = 440.26, neutral: mean = 755.85ms, SD = 376.74ms).

The late bilinguals showed the reverse pattern, demonstrating a slower reaction time in the emotion condition in English (mean = 542.31, SD = 255.28) compared to the neutral condition in English (mean = 515.77ms, SD = 196.49ms), but the same pattern as the early bilinguals across the French conditions, such that they performed slightly faster in the French emotion condition (mean = 533.33ms, SD = 215.51) than the French neutral condition (mean = 536.77ms, SD = 204.00ms), but to a lesser extent than the early bilinguals demonstrated.

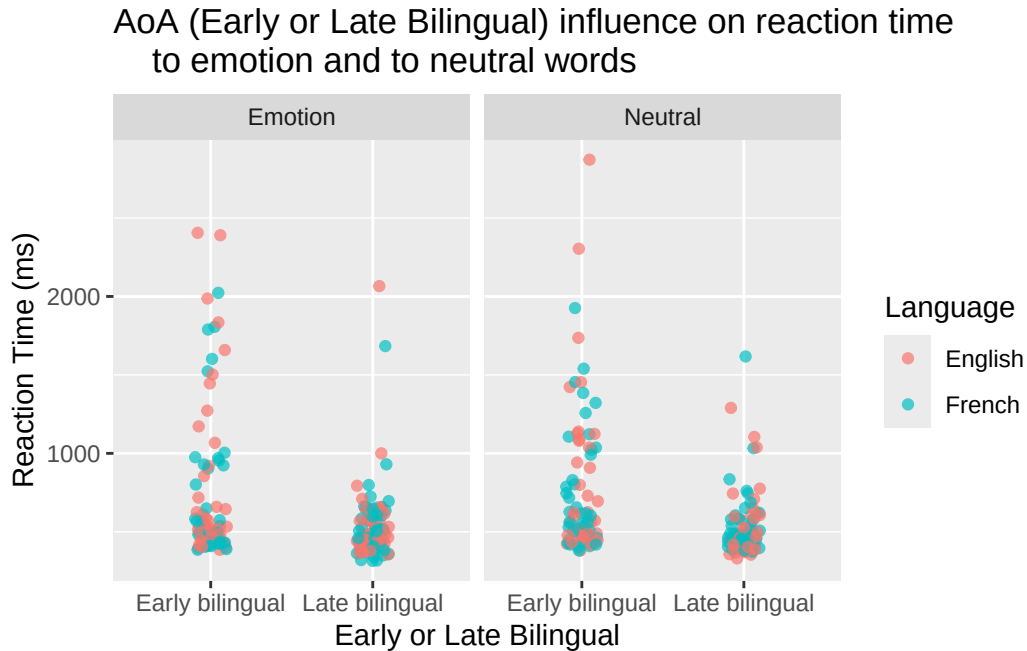
Taken together, late bilinguals had faster reaction times than early bilinguals, contrary to predictions, with early bilinguals perhaps showing an emotional interference effect in the French but not the English condition (where they performed slightly faster in English emotion than

English neutral words), and late bilinguals perhaps showing an emotional interference effect in the English condition but not the French condition (where they performed slightly faster in French emotional than French neutral words).

The differing patterns across participant groups in the raw data therefore suggest that the emotional interference effect may differ with age of acquisition, such that late bilinguals overall showed a faster reaction time across both language and condition, suggesting a lack of emotional interference effect, particularly on the French emotion condition, which is their L2. However, raw data results are rather varied, and suggest that any emotional interference effect was not consistent, for example early bilinguals had a stronger interference effect in the neutral condition than the emotion condition. Statistical analysis is required to verify these patterns.

0.3.3 Plotting the raw data

```
newfinaldata |>
  ggplot(aes(early_late, RT, colour = Language)) +
  geom_jitter(alpha = 0.7, width = 0.1) +
  facet_wrap(~ Condition) +
  labs(
    x = "Early or Late Bilingual",
    y = "Reaction Time (ms)",
    title = "AoA (Early or Late Bilingual) influence on reaction time
to emotion and to neutral words")
```



0.3.3.1 Figure 1 - Age of acquisition (early or late bilingual) effect on reaction time (RT) across condition (emotion and neutral stimuli) and language (French and English).

I used a jitter plot to represent the raw data, since it combines numeric (reaction times) and categorical variables. The jitter plot is faceted by Condition (neutral and emotion stimuli), with the x-axis showing Participant group (early or late bilingual) and the y-axis showing Reaction Time (ms). The colours differentiate the Language of the stimuli (French = blue, English = red).

The plot shows firstly in the emotion condition, that late bilinguals showed less variation and therefore are more densely represented than the early bilinguals. There is minimal variation across both participant groups between languages (English compared to French).

In the neutral condition, there is slightly more variation across languages, with late bilinguals appearing to perform very slightly faster in English than in French, and early bilinguals' reaction times showing more variation compared to late bilinguals in this condition. The plot reflects a slightly faster reaction time across conditions and across language for late bilinguals.

There are some outliers across condition, group and language, but these appear to be more pronounced in the early bilinguals compared to the late bilinguals.

0.3.4 Regression modelling

I used Bayesian log-normal regression model for the outcome variable reaction time (RT) using brms, as this distribution allows positive-only values, continuous variables, and is able to account for right-skewed data. This was fitted with Group (early vs late bilinguals), Language (French vs English) and Condition (emotion vs neutral) and their interactions as between-participant fixed effects, and Language (French vs English) and Condition (emotion vs neutral) as within-participant varying slopes, and including Participant ID as by-subject varying intercept to account for inter-participant variability. I then obtained summary measures. As the values are reported as logged reaction time, I used `exp()` to unlog on the relevant values.

```
brm_newfinaldata <- brm(
  RT ~ early_late * Language * Condition +
    (Language + Condition | id ),
  family = lognormal,
  data = newfinaldata,
  seed = 456,
  cores = 4,
  file = "cache/brmfinaldata",
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)
```

```
summary(brm_newfinaldata)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~id (Number of levels: 11)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat
sd(Intercept)	0.40	0.12	0.24	0.69	1.00
sd(LanguageFrench)	0.05	0.04	0.00	0.15	1.00
sd(ConditionNeutral)	0.06	0.05	0.00	0.18	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.94	0.70	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.86	0.79	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.89	0.85	1.00
	Bulk_ESS	Tail_ESS			
sd(Intercept)	1410	1785			

sd(LanguageFrench)	1939	1869
sd(ConditionNeutral)	1508	1335
cor(Intercept,LanguageFrench)	3895	2755
cor(Intercept,ConditionNeutral)	4083	2955
cor(LanguageFrench,ConditionNeutral)	2487	2824

Regression Coefficients:

	Estimate	Est.Error	
Intercept	6.54	0.18	
early_lateLatebilingual	-0.29	0.25	
LanguageFrench	-0.08	0.07	
ConditionNeutral	0.01	0.07	
early_lateLatebilingual:LanguageFrench	0.07	0.09	
early_lateLatebilingual:ConditionNeutral	-0.05	0.09	
LanguageFrench:ConditionNeutral	0.07	0.08	
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12	
	1-95% CI	u-95% CI	Rhat
Intercept	6.17	6.92	1.00
early_lateLatebilingual	-0.79	0.18	1.01
LanguageFrench	-0.21	0.05	1.00
ConditionNeutral	-0.12	0.14	1.00
early_lateLatebilingual:LanguageFrench	-0.11	0.25	1.00
early_lateLatebilingual:ConditionNeutral	-0.23	0.13	1.00
LanguageFrench:ConditionNeutral	-0.10	0.24	1.00
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.24	0.23	1.00
	Bulk_ESS	Tail_ESS	
Intercept	965	1898	
early_lateLatebilingual	981	1825	
LanguageFrench	2101	2546	
ConditionNeutral	2164	2434	
early_lateLatebilingual:LanguageFrench	2167	2787	
early_lateLatebilingual:ConditionNeutral	2229	2620	
LanguageFrench:ConditionNeutral	2156	2769	
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687	

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.26	0.30	1.00	5596	3054

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
exp(6.17) #Intercept_95%_lower_bound
```

```
[1] 478.1861
```

```
exp(6.92) #Intercept_95%_upper_bound
```

```
[1] 1012.32
```

The model did converge, however the summary measures reflect a high degree of uncertainty.

At 95% credible interval, the model estimates that the mean reaction time for early bilinguals in the emotion and English condition is between 6.17 (lower bound) and 6.92 (upper bound) logged milliseconds, and 478.19ms (lower bound) and 1,0312.32ms (upper bound) unlogged milliseconds. All predictors cross zero at 95% credible interval, and therefore there is a high degree of uncertainty as to their effect size.

```
summary(brm_newfinaldata, prob = 0.8)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~id (Number of levels: 11)

	Estimate	Est.Error	1-80% CI	u-80% CI	Rhat
sd(Intercept)	0.40	0.12	0.28	0.55	1.00
sd(LanguageFrench)	0.05	0.04	0.01	0.11	1.00
sd(ConditionNeutral)	0.06	0.05	0.01	0.13	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.85	0.37	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.71	0.53	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.71	0.63	1.00

	Bulk_ESS	Tail_ESS
sd(Intercept)	1410	1785
sd(LanguageFrench)	1939	1869
sd(ConditionNeutral)	1508	1335
cor(Intercept,LanguageFrench)	3895	2755
cor(Intercept,ConditionNeutral)	4083	2955


```
cor(LanguageFrench,ConditionNeutral)      2487      2824
```

Regression Coefficients:

	Estimate	Est.Error			
Intercept	6.54	0.18			
early_lateLatebilingual	-0.29	0.25			
LanguageFrench	-0.08	0.07			
ConditionNeutral	0.01	0.07			
early_lateLatebilingual:LanguageFrench	0.07	0.09			
early_lateLatebilingual:ConditionNeutral	-0.05	0.09			
LanguageFrench:ConditionNeutral	0.07	0.08			
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12			
	1-80% CI	u-80% CI	Rhat		
Intercept	6.31	6.76	1.00		
early_lateLatebilingual	-0.59	0.01	1.01		
LanguageFrench	-0.16	0.00	1.00		
ConditionNeutral	-0.08	0.09	1.00		
early_lateLatebilingual:LanguageFrench	-0.05	0.18	1.00		
early_lateLatebilingual:ConditionNeutral	-0.17	0.07	1.00		
LanguageFrench:ConditionNeutral	-0.04	0.17	1.00		
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.15	0.14	1.00		
	Bulk_ESS	Tail_ESS			
Intercept	965	1898			
early_lateLatebilingual	981	1825			
LanguageFrench	2101	2546			
ConditionNeutral	2164	2434			
early_lateLatebilingual:LanguageFrench	2167	2787			
early_lateLatebilingual:ConditionNeutral	2229	2620			
LanguageFrench:ConditionNeutral	2156	2769			
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687			

Further Distributional Parameters:

	Estimate	Est.Error	1-80% CI	u-80% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.26	0.29	1.00	5596	3054

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat = 1`).

```
exp(6.31) #Intercept_80%_lower_bound
```

```
[1] 550.0449
```

```
exp(6.76) #Intercept_80%_upper_bound
```

```
[1] 862.6422
```

At 80% credible interval, all predictors still cross zero, but to a lesser extent. The mean reaction time predicted at the Intercept (english_emotion_early) is between 6.31 logged milliseconds and 6.76 logged milliseconds, and 550.04 and 862.64 unlogged milliseconds.

```
summary(brm_newfinaldata, prob = 0.6)
```

```
Family: lognormal
Links: mu = identity
Formula: RT ~ early_late * Language * Condition + (Language + Condition | id)
Data: newfinaldata (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~id (Number of levels: 11)

	Estimate	Est.Error	1-60% CI	u-60% CI	Rhat
sd(Intercept)	0.40	0.12	0.31	0.49	1.00
sd(LanguageFrench)	0.05	0.04	0.02	0.09	1.00
sd(ConditionNeutral)	0.06	0.05	0.02	0.10	1.00
cor(Intercept,LanguageFrench)	-0.32	0.46	-0.74	0.09	1.00
cor(Intercept,ConditionNeutral)	-0.12	0.45	-0.55	0.30	1.00
cor(LanguageFrench,ConditionNeutral)	-0.07	0.49	-0.54	0.40	1.00
	Bulk_ESS	Tail_ESS			
sd(Intercept)	1410	1785			
sd(LanguageFrench)	1939	1869			
sd(ConditionNeutral)	1508	1335			
cor(Intercept,LanguageFrench)	3895	2755			
cor(Intercept,ConditionNeutral)	4083	2955			
cor(LanguageFrench,ConditionNeutral)	2487	2824			

Regression Coefficients:

	Estimate	Est.Error
Intercept	6.54	0.18
early_lateLatebilingual	-0.29	0.25
LanguageFrench	-0.08	0.07
ConditionNeutral	0.01	0.07
early_lateLatebilingual:LanguageFrench	0.07	0.09

early_lateLatebilingual:ConditionNeutral	-0.05	0.09	
LanguageFrench:ConditionNeutral	0.07	0.08	
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.01	0.12	
	1-60% CI	u-60% CI	Rhat
Intercept	6.39	6.67	1.00
early_lateLatebilingual	-0.49	-0.09	1.01
LanguageFrench	-0.14	-0.03	1.00
ConditionNeutral	-0.05	0.06	1.00
early_lateLatebilingual:LanguageFrench	-0.01	0.14	1.00
early_lateLatebilingual:ConditionNeutral	-0.13	0.03	1.00
LanguageFrench:ConditionNeutral	-0.00	0.13	1.00
early_lateLatebilingual:LanguageFrench:ConditionNeutral	-0.10	0.09	1.00
	Bulk_ESS	Tail_ESS	
Intercept	965	1898	
early_lateLatebilingual	981	1825	
LanguageFrench	2101	2546	
ConditionNeutral	2164	2434	
early_lateLatebilingual:LanguageFrench	2167	2787	
early_lateLatebilingual:ConditionNeutral	2229	2620	
LanguageFrench:ConditionNeutral	2156	2769	
early_lateLatebilingual:LanguageFrench:ConditionNeutral	2232	2687	

Further Distributional Parameters:

	Estimate	Est.Error	1-60% CI	u-60% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.28	0.01	0.27	0.29	1.00	5596	3054

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
exp(6.39) #Intercept_60%_lower_bound
```

```
[1] 595.8566
```

```
exp(6.67) #Intercept_60%_upper_bound
```

```
[1] 788.3956
```

```
exp(-0.49) #early_lateLatebilingual_60%_lower_bound
```

```
[1] 0.6126264
```

```
exp(-0.09) #early_lateLatebilingual_60%_upper_bound
```

```
[1] 0.9139312
```

```
exp(-0.14) #LanguageFrench_60%_lower_bound
```

```
[1] 0.8693582
```

```
exp(-0.03) #LanguageFrench_60%_upper_bound
```

```
[1] 0.9704455
```

Finally, at 60% credibility, the predicted reaction time at the intercept is between 6.17 and 6.92 logged milliseconds, or between 595.86 and 788.38 unlogged milliseconds (english_emotion_early). At 60% credibility, two predictors now show a decrease in log milliseconds. Firstly, english_emotion_late shows an effect size between -0.49 and -0.09 (logged) or 0.61 to 0.91 (unlogged), suggesting that late bilinguals are predicted to be faster than early bilinguals in the emotion/English condition. Secondly, french_emotion_early show an effect size between -0.14 and -0.03 (logged) or 0.86 to 0.97 (unlogged), suggesting that early bilinguals in the emotion condition are likely to show a faster reaction time in French than in English. However, the remaining predictors cross zero and therefore there remains a high degree of uncertainty regarding how these predictors interact across the model.

After obtaining summary measures at 95%, 80% and 60% credible intervals, I obtained the model draws, implementing the relevant predictors and interactions based on the b_Intercept as english_emotion_early. I then used select() the relevant columns (predictors and their interactions) and pivot_longer() in order to make new columns for these predictors, which I could then plot into the graph (below).

```
draws_brm_newfinaldata <- as_draws_df(brm_newfinaldata)

frame_draws_brm_newfinaldata <- draws_brm_newfinaldata |>
  mutate(
    eng_emotion_early = b_Intercept,
    fr_emotion_early = b_Intercept + b_LanguageFrench,
    eng_emotion_late = b_Intercept + b_early_lateLatebilingual,
    eng_neutral_early = b_Intercept + b_ConditionNeutral,
    fr_emotion_late = b_Intercept + b_early_lateLatebilingual
    + b_LanguageFrench + `b_early_lateLatebilingual:LanguageFrench`,
    eng_neutral_late = b_Intercept + b_ConditionNeutral +
```

```

    b_early_lateLatebilingual +
    `b_early_lateLatebilingual:ConditionNeutral`,
fr_neutral_early = b_Intercept + b_LanguageFrench +
    b_ConditionNeutral +
    `b_LanguageFrench:ConditionNeutral`,
fr_neutral_late = b_Intercept + b_ConditionNeutral +
    b_LanguageFrench +
    b_early_lateLatebilingual +
    `b_early_lateLatebilingual:LanguageFrench` +
    `b_LanguageFrench:ConditionNeutral` +
    `b_early_lateLatebilingual:ConditionNeutral` +
    `b_early_lateLatebilingual:LanguageFrench:ConditionNeutral`
) |>
select(eng_emotion_early, fr_emotion_early, eng_emotion_late,
       eng_neutral_early, fr_emotion_late, eng_neutral_late,
       fr_neutral_early, fr_neutral_late)

```

Warning: Dropping 'draws_df' class as required metadata was removed.

```

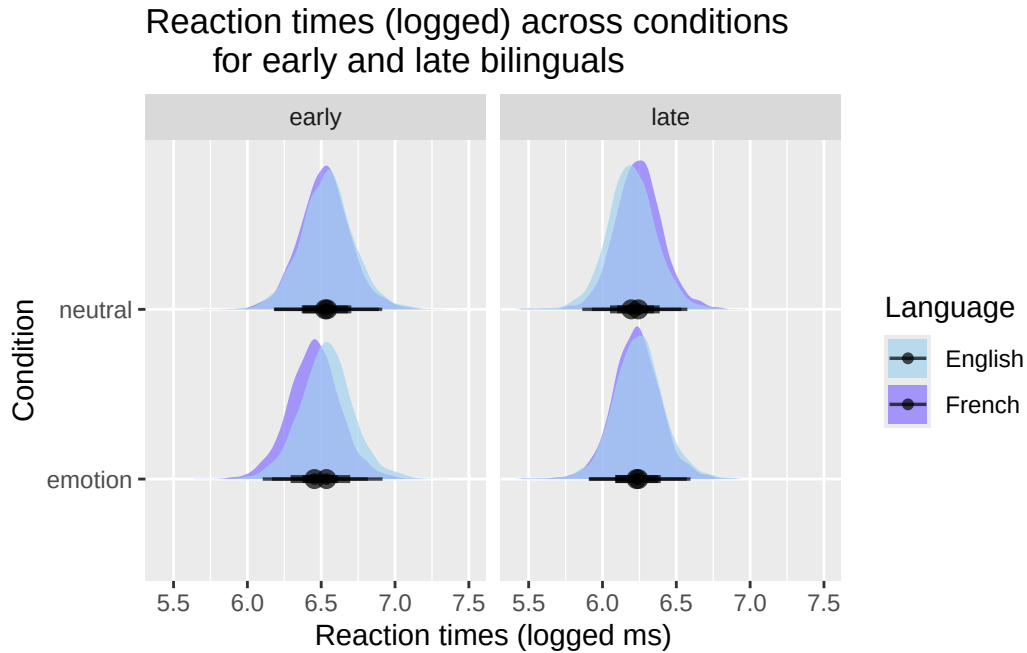
plot2 <- frame_draws_brm_newfinaldata |>
  select(eng_emotion_early, fr_emotion_early, eng_emotion_late,
         eng_neutral_early, fr_emotion_late, eng_neutral_late,
         fr_neutral_early, fr_neutral_late) |>
  pivot_longer(
    eng_emotion_early:fr_neutral_late,
    names_to = "cond",
    values_to = "epred"
  ) |>
  separate(cond, c("Language", "Condition", "Early_Late"))

```

```

plot2 |>
  ggplot(aes(epred, Condition, fill = Language )) +
  stat_halfeye(alpha = 0.7) +
  scale_fill_manual(
    labels = c("English", "French"),
    values = c("lightskyblue2", "slateblue1")
  ) + facet_grid(cols = vars(Early_Late)) +
  labs(title = "Reaction times (logged) across conditions
         for early and late bilinguals", x = "Reaction times (logged ms)")

```



0.3.4.1 Figure 2 - Age of acquisition (early or late bilingual) effect on logged reaction time (RT) across condition (emotion and neutral stimuli) and language (French and English).

I used a half-eye plot to demonstrate logged Reaction times on the x-axis, faceted by Participant group (early, late), and Condition on the y-axis (neutral, emotion), with colour differentiating language (English, French). The plot reflects the model draws, showing very little variability between performance of early and late bilinguals across condition and across language. The most noticeable variation is in the emotion condition for early bilinguals, who appear to show a slightly faster reaction time in French than in English: the model did predict a negative effect on reaction time (i.e. faster reaction time) in these conditions at 60% credibility, however the summary measures showed this to still have a large degree of uncertainty.

0.4 Discussion

The present study investigated if age of acquisition influences reaction time to French and English emotion compared to neutral words in a bilingual Stroop paradigm, comparing the emotional interference effect across early and late bilinguals groups and across languages (French compared to English).

Hypothesis 1 predicted that an emotional interference effect would be observed, with both groups showing slower reaction time in the emotion condition compared to the neutral condition. In terms of the overall emotional interference effect, the statistical analysis did not

confirm an effect, since the results were highly variable, and therefore no strong conclusions can be drawn.

Hypothesis 2 and 3 predicted that early bilinguals would display an equal amount of emotional interference across both languages, meaning similar reaction times will be shown, while late bilinguals would show an emotional interference effect (slower reaction time) only in the English condition, as the emotional interference effect is expected to be stronger in the L1.

While the statistical analysis returned highly uncertain results, at 60% credibility it can be predicted that late bilinguals perform faster than early bilinguals in the emotion condition in English, suggesting that an emotional interference effect may be greater for early bilinguals than late bilinguals in English. Secondly, also at 60% credibility, early bilinguals are likely to show a faster reaction time in French than in English, suggesting a greater interference effect in English than in French for the early bilinguals. Taken together, this suggests that early bilinguals may show a greater emotional interference effect in English (compared to French, and compared to late bilinguals), which is contrary to hypothesis 2, that they would show an equal interference effect across languages.

However, due to the high uncertainty of results, no conclusions regarding Hypothesis 1, 2, or 3 can be drawn at this stage. In particular, the interactions between age of acquisition, conditions and languages were uncertain, meaning the specific emotional interference effect in L1 compared to L2 is uncertain. Overall therefore, in regards to the research question, the study returned inconclusive results regarding how age of acquisition interacts with the emotional interference effect across L1 and L2.

0.4.1 Limitations

There are several limitations to the study which could explain the uncertainty of the results.

Firstly, it is still a small participant cohort (5 early bilinguals and 6 late bilinguals), so including more participants may have returned a higher degree of certainty in the results.

Secondly, operationalising age of acquisition as a categorical variable may have obscured between-group differences (Singleton and Pfenninger 2022; Wagner, Bialystok, and Grundy 2022), and therefore treating age of acquisition as a continuous variable may have led to different results.

Additionally, since the main focus of the study was investigating the emotional Stroop effect, rather than the Stroop effect associated with orthographical/colour interference, I chose to include the incorrect trials alongside the correct trials. Removing the incorrect trials may have returned more certainty.

Furthermore, while this study investigated age of acquisition as a predictor of the emotional interference effect across languages, studies reflect that processing of emotion words in L1 and L2 may vary with proficiency, with lower proficiency typically leading to a greater interference

effect (Sutton et al. 2007). As proficiency in L2 French was not controlled in this study, age of acquisition may have been confounded with proficiency, in particular since the late bilingual group also had a lower proficiency level.

Finally, it is likely that the stimuli presented some difficulties, since the stimuli used by Sutton et al. (2007) and replicated in this study contain uncommon lexical items (nautical nouns), which are uncommon both in English and in French. It is possible that participants were surprised by these stimuli, meaning that any effect of emotional interference was obscured by a processing delay of encountering an uncommon noun. The late bilingual group, who had lower levels of proficiency in French than the early bilingual group, may also have been completely unfamiliar with these neutral lexical items, which may have conversely affected their reaction times compared to a different set of more common neutral lexical items. It is clear that further investigation is required.

0.5 Appendix

0.5.1 Stimuli

English - Emotion	French - Emotion	English - Neutral	French - Neutral
nervous	nerveux	engine	barre
grief	chagrin	boom	bôme
upset	contrarié	stern	poupe
greed	avidité	cabin	cabine
anxiety	anxiété	keel	quille
jealousy	jalousie	motor	moteur
lonely	seul	seat	siège
scared	effrayé	deck	pont
afraid	craintif	driver	conducteur
panic	panique	propeller	hélice
fear	peur	floor	plancher
angry	énervé	anchor	ancre
disgusted	dégoûté	paddle	pagaie
depressed	déprimé	beam	barrot
mad	furieux	port	bâbord
troubled	tourmenté	rope	corde

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